



AGUTS

Ahmedabad Gandhinagar Unified Transit System

A Multi-Modal Public Transit Information System

Metro Rail · BRTS · AMTS · GSRTC

Metro (GMRC)

BRTS – Janmarg

AMTS

GSRTC

Project Information

Course IT-214 Database Management System

Lab Group 5

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Project Title and Overview

Project Name: Ahmedabad Gandhinagar Unified Transit System (AGUTS)

Domain: Public Urban Transportation – Multi-Modal Transit System

Geographic Scope: Ahmedabad and Gandhinagar, Gujarat, India

Project Description

The **Ahmedabad Gandhinagar Unified Transit System (AGUTS)** is a comprehensive database system designed to serve as the central information backbone for the multi-modal public transportation network spanning the twin cities of **Ahmedabad** and **Gandhinagar** in Gujarat, India.

The Ahmedabad–Gandhinagar corridor is one of India’s fastest-growing urban regions, home to millions of daily commuters who rely on a mix of Metro Rail, BRTS, AMTS, and GSRTC services to travel for work, education, and essential needs. Despite this scale, each transit operator maintains its data independently – resulting in fragmented schedules, inconsistent fare information, and no single platform where a commuter can plan a multi-modal journey, check real-time availability, or track their travel history. AGUTS is designed to unify this fragmented data landscape into a single, coherent database system that serves commuters, operators, and administrators alike.

Transit Services Covered

- **Ahmedabad Metro (GMRC)** – Metro rail service covering Line 1 (Vastrapur ↔ Thaltej Gam) and Line 2 (Motera/Gyaspur ↔ GNLU/Mahatma Mandir, Gandhinagar). Token-based and smart-card AFC ticketing.
- **BRTS – Janmarg** – Bus Rapid Transit System on dedicated corridors across Ahmedabad, operated by Janmarg Ltd. Closed stations, prepaid ticketing, 20+ routes.
- **AMTS** – Ahmedabad Municipal Transport Service covering city and suburban routes in Ahmedabad and Gandhinagar. Conductor-based ticketing with 150+ routes.
- **GSRTC** – Gujarat State Road Transport Corporation providing inter-city and Gandhinagar-specific bus services connecting the two cities and beyond.

Key Functional Areas

Functional Area	Description
Network & Infrastructure	Routes, stops, stations, Metro lines, interchange points, GPS coordinates
Schedules & Trips	Timetables, trip assignments, stop-wise departure times, frequency
Fleet Management	Vehicle registration, type, capacity, fuel type, maintenance history
Staff Management	Drivers, conductors, motormen – assignments, licences, shifts
Passenger Management	Registration, smart cards, travel history, concession categories
Ticketing & Passes	Tokens, daily/weekly/monthly passes, smart-card balance, transactions
Fares & Revenue	Fare tables, zone/distance-based pricing, inter-modal transfer fares
Complaints & Feedback	Passenger complaints, resolution tracking, service ratings
Reporting & Analytics	Ridership stats, revenue, punctuality, fleet-utilisation reports

Application Users and Use Cases

- **Guest** – Search routes, view schedules and fares without logging in.
- **Registered Passenger** – Manage smart card and passes, view travel history, file complaints, rate services.
- **Ticketing Agent** – Issue tokens, smart cards, and passes; record transactions.
- **Driver / Conductor / Motorman** – View assigned trips, log completion, report vehicle issues.
- **Depot / Station Manager** – Assign vehicles and staff, schedule maintenance, view punctuality reports.
- **Transit Authority Admin** – Manage routes, stops, fares, fleet, staff; view ridership and revenue reports.

Registration Details

Passenger: Name, Email, Phone, Date of Birth, City, Concession category (General / Student / Senior / Differently-Abled). A unique Passenger ID is assigned on registration.

Staff: Registered by their operator. Details: Employee ID, Name, Role, Date of joining, Contact, Home depot, and role-specific credentials (driving licence / metro licence).

Business Rules and Scope

- A smart card belongs to exactly one passenger.
- A trip must have one vehicle and one staff member assigned.
- A pass is valid for one operator only.
- A vehicle can be assigned to a trip only if its status is *Active*.

Out of Scope: Real-time GPS tracking, online payments, and mobile ticketing.

Tentative Query List

General:

1. Routes between two stops.
2. Fare between stops by operator and concession category.
3. Stop sequence and timings for a route.

Passenger:

1. My travel history and transactions.
2. Smart card balance and active passes.
3. Status of my complaints.

Operator / Admin:

1. Ridership on a route for a given date range.
2. Trips delayed by more than 10 minutes.
3. Revenue by operator and period.
4. Vehicles currently under maintenance.
5. Open complaints for an operator.
6. Trips assigned to a specific driver or motorman.

Why This Project?

Public transit in Ahmedabad and Gandhinagar involves multiple independent operators – Metro, BRTS, AMTS, and GSRTC – each managing their own data in isolation. A commuter travelling across services has no unified source to check schedules, compare fares, or plan a multi-modal journey. Operators themselves lack cross-system visibility into ridership, fleet utilization, and service performance.

AGUTS addresses this by providing a single, well-structured database that captures all transit data – routes, stops, trips, vehicles, staff, passengers, tickets, and fares – in one place. This enables commuters to get consistent information, operators to manage their services efficiently, and administrators to make data-driven decisions about the transit network as a whole.

Project Milestones

#	Milestone
1	Scenario Description (<i>this document</i>)
2	ER Diagram (Dia) – cardinality & participation constraints
3	Relational Schema, BCNF Normalization Proofs, DDL Scripts
4	SQL Retrieval Queries (10–15) + Sample Data (INSERT scripts)
5	Console Application (Java JDBC or Embedded SQL in C)
6	Final Document
7	Final Presentation & Viva